

Reproducing *Physics of Language Models*: Part 1, Context-Free Grammar

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Abstract

We present a partial reproduction of Allen-Zhu and Li [1] (*Physics of Language Models: Part 1, Context-Free Grammar*), which studies how transformer language models learn the latent hierarchical structure of context-free grammars (CFGs). We re-implement the training and evaluation pipelines, train a GPT-2 model with rotary positional embeddings (RoPE) on the `cfg3b` synthetic grammar for 6,500 steps, and evaluate it through completion accuracy, KL divergence, linear probing of hidden states, and attention-pattern analysis. Our results largely confirm the qualitative findings of the original paper, while quantitative gaps—particularly in KL divergence—are attributable to the reduced training budget.

1 Introduction

A central question in the study of large language models is whether next-token prediction alone is sufficient for a model to discover the latent structure of the language it is trained on. Allen-Zhu and Li [1] investigate this in a controlled setting: synthetic languages generated by context-free grammars (CFGs), where ground-truth parse trees are available by construction. This allows precise, quantitative questions to be asked about the model’s internal representations.

We reproduce their experimental setup using a standard GPT-2 architecture. Among the synthetic grammars introduced by Allen-Zhu and Li [1], we focus on `cfg3b`, the easiest of the family: under computational constraints, its lower complexity makes it the most tractable grammar to learn within a limited training budget, while still exhibiting the deep hierarchical structure central to the paper’s analysis. Our goal is to verify whether the original conclusions hold under a reduced training budget (6,500 vs. the paper’s 100,000 iterations).

2 Background

2.1 Context-Free Grammars as Synthetic Languages

A context-free grammar $G = (V_N, V_T, R, S)$ defines a language over terminal symbols V_T via rewriting rules in R . Each string $x = (x_1, \dots, x_n)$ is derived from the start symbol S through L levels of non-terminal expansions. At each level ℓ , every terminal position i has a unique ancestor non-terminal symbol $s_\ell(i)$ and ancestor index $p_\ell(i)$ (the ordinal position of that non-terminal among all level- ℓ nodes in the parse tree, left to right). A *boundary* at level ℓ and position i is defined as:

$$b_\ell(i) = \mathbf{1}[p_\ell(i) \neq p_\ell(i+1)],$$

i.e., position i is the last terminal of a constituent at level ℓ .

The grammar `cfg3b` contains 32 production rules over 16 non-terminal symbols and 3 terminal symbols, with a hierarchy exactly 7 levels deep and branching factors of 2 or 3. Its level structure is shown in Figure 1. The exact rules are excluded for readability but can be found in the original Allen-Zhu and Li [1].

2.2 Probing Hidden States

Following Allen-Zhu and Li [1], we use *multi-head linear probing* to test whether the hidden state $h_i \in \mathbb{R}^d$ at position i encodes structural information. A probe $f_\theta(h_i)$ is trained to predict the ancestor symbol $s_\ell(i)$.

Two probe variants are used:

Full probe (R4). Unrestricted attention; accuracy evaluated over all positions.

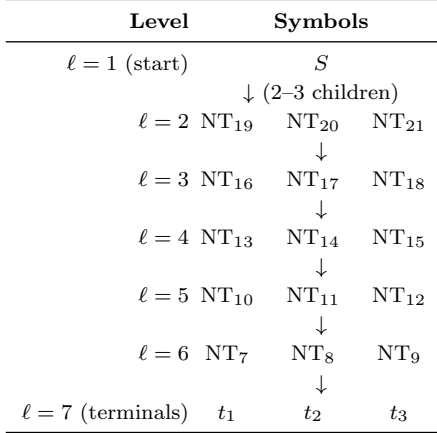


Figure 1: Hierarchical structure of **cfg3b**. Levels 2–6 each contain exactly 3 non-terminals; each rewrites to 2 or 3 children from the level below. These NT groups are the probe targets in Table 2.

Diagonal probe (R5). Attention restricted to a band of width $2\delta + 1$ around the diagonal; accuracy evaluated at boundary positions only.

3 Methods

3.1 Model and Training

We train a GPT-2 model [2] with rotary positional embeddings (RoPE, **gpt_rot**): 12 layers, 12 heads, embedding dimension $d = 768$, sequence length 512, vocabulary size 5 ($< BOS >$, 1, 2, 3, $< EOS >$).

Training uses AdamW ($\beta_1=0.9$, $\beta_2=0.98$, weight decay 0.1, learning rate 3×10^{-4}) with a linear decay schedule over $N=6,500$ steps. We use gradient accumulation over 8 micro-batches of size 12 (effective batch size 96). Strings are sampled on-the-fly from **cfg3b**.

3.2 Evaluation

We evaluate the model on four tasks:

Completion accuracy (R1). The model generates a completion autoregressively from a prefix of length 0 or 50. A completion is correct if the full string is a valid string of the grammar. Reported under both multinomial sampling and greedy decoding.

KL divergence (R3). We compute $\text{KL}(P_{\text{true}} \| P_{\text{model}})$ averaged over positions across 200 sampled strings.

Probing (R4–R5). For each non-terminal level ℓ , we train a multi-head linear probe for 5,000 iterations

on the frozen hidden states of the last transformer block.

Attention analysis (R6–R9). We measure how attention is distributed relative to token distance and constituent boundaries.

4 Results

4.1 Completion Accuracy and KL Divergence

Table 1 summarises our quantitative results on **cfg3b** after 6,500 training steps.

Table 1: Quantitative results for **gpt_rot** on **cfg3b** (6,500 steps).

Metric	Value
Completion acc. (multinomial, no prefix)	98.00%
Completion acc. (multinomial, prefix = 50)	99.50%
KL divergence (nats/token)	0.000350
Paper target (GPT_rot, cfg3b)	0.000090

Multinomial completion accuracy is high despite the reduced training budget: 98.00% from scratch and 99.50% when conditioned on a 50-token prefix. This indicates that the model has already learned to assign probability mass over valid continuations effectively at 6,500 steps.

The KL divergence of 0.000350 nats/token is about $4\times$ above the paper’s target of ≈ 0.00009 for a fully trained model on the same grammar. This gap is consistent with early-stage training: the model has learned a rough approximation of the distribution but has not yet fully converged to the true grammar distribution.

4.2 Probing (Results 4–5)

We train multi-head linear probes on the frozen hidden states of the last transformer block for each non-terminal level $\ell \in \{2, \dots, 6\}$ of **cfg3b**. Table 2 reports full-probe accuracy (R4, evaluated over all positions) and diagonal-probe accuracy (R5, evaluated only at boundary positions with $\delta = 0$).

Full-probe accuracy (R4) improves at deeper levels (closer to the terminal layer), reaching 87.1% at NT6. This is consistent with lower-level constituents being smaller and more locally decodable. The shallower levels—NT2 and NT3—show weaker accuracy (53.3% and 59.1%), which we attribute to insufficient training: at 6,500 steps the model has not yet fully encoded higher-level constituency.

Table 2: Probing accuracy for `gpt_rot` on `cfg3b` (6,500 steps) vs. a fully-trained `GPT_rot` on the same grammar [1].

Level	Full (R4)		Diagonal (R5)	
	Ours	Paper	Ours	Paper
NT2	53.3%	100%	87.1%	100%
NT3	59.1%	100%	66.3%	99.8%
NT4	78.0%	100%	89.5%	99.6%
NT5	86.1%	100%	99.9%	100%
NT6	87.1%	100%	91.9%	100%

The diagonal probe (R5) consistently and substantially outperforms the full probe trained for the same number of iterations. The gap is striking: at NT5, R5 reaches 99.9% versus R4’s 86.1%; at NT2, R5 reaches 87.1% while R4 only reaches 53.3%. This performance advantage, achieved with identical compute, directly supports the original paper’s central claim: *the hidden states at NT-boundary positions already encode the full ancestry information of the constituent they close*. Because the diagonal probe is restricted to a narrow window around position i , its high accuracy means the relevant information is locally available at boundary positions without needing global context. At non-boundary positions, the hidden state is consistent with multiple possible subtrees, making classification information-theoretically impossible—a result formalised in the original paper.

4.3 Attention Analysis (Results 6–9)

Result 6 — Distance-based attention. Averaging attention weights $\bar{A}_p$ over all pairs (i, j) with $j - i = p$, across all layers and heads, we observe a strong recency bias. The distribution peaks at $p = 1$ ($\bar{A} = 0.0946$) and $p = 2$ ($\bar{A} = 0.0927$), decaying rapidly thereafter (e.g. $\bar{A} = 0.0097$ at $p = 15$). A uniform distribution over 100 positions would give $\bar{A} \approx 0.01$, so the model attends roughly $9\times$ more to the immediately preceding token than baseline. This distance bias is subtracted to isolate structure-sensitive attention.

Result 7 — Attention to NT-boundaries. After removing the distance bias, the residual $B = A - \bar{A}$ is measured for key positions at offset δ from a constituent boundary. Table 3 shows a clear peak at $\delta = 0$ across all levels: the model disproportionately attends to exact boundary positions. The effect is strongest at NT2 ($B = +0.0085$) where boundaries are rare and summarise large subtrees, decreasing toward NT6 ($B = +0.0019$) where boundaries are

frequent and subtrees are small.

Table 3: Result 7 — Residual attention B at offset δ from NT-boundary

	$\delta=-2$	$\delta=-1$	$\delta=0$	$\delta=+1$	$\delta=+2$
NT2	-.0008	-.0018	+.0085	-.0024	-.0013
NT3	-.0006	-.0017	+.0065	-.0019	-.0012
NT4	-.0005	-.0016	+.0064	-.0018	-.0009
NT5	-.0003	-.0015	+.0047	-.0006	-.0010
NT6	+.0006	-.0016	+.0019	-.0008	-.0002

Result 8 — Boundary-to-boundary attention.

Restricting both query and key positions to NT-boundaries at the same level, the mean residual B is positive throughout the hierarchy (Table 4), peaking at NT4 where boundary density and subtree size are best balanced for informative pairings.

Table 4: Result 8 — mean residual B for same-level NT-boundary pairs

	NT2	NT3	NT4	NT5	NT6
mean B	+.0020	+.0056	+.0080	+.0059	+.0020

Result 9 — Adjacent NT-end attention.

Among same-level boundary pairs, we bin B by ancestor distance $r = p_\ell(j) - p_\ell(i)$ (the $r=0$ column is empty by construction, as a level- ℓ subtree has a single boundary). Table 5 shows the signal peaks sharply at $r = 1$ at every level and decays outward, with the strongest residuals of the whole analysis at NT4–NT5 ($+0.0298$, $+0.0315$). Far- r buckets are sparsely sampled and approach noise (occasional small negatives), and NT2 has too few high-level constituents to register beyond $r = 1$. This monotone decay is the signature of a DP-like recurrence: the model retrieves structural information primarily from the immediately preceding same-level boundary, mirroring CYK-style parsing.

Table 5: Result 9 — residual B by ancestor distance r between same-level boundaries.

	$r=1$	$r=2$	$r=3$	$r=4$	$r=5$
NT2	+.0020	.0000	.0000	.0000	.0000
NT3	+.0122	+.0023	+.0011	-.0003	.0000
NT4	+.0298	+.0117	+.0063	+.0068	+.0036
NT5	+.0315	+.0211	+.0038	+.0054	+.0052
NT6	+.0204	+.0139	+.0022	+.0020	-.0008

5 Extension: Present vs. Used Information

5.1 Motivation and Target Level

The probing experiments (R4–R5) establish that NT ancestry information is *present* in the model’s hidden states. Our extension asks a sharper question: is this information also *used* by the model when generating subsequent tokens? We investigate this via activation patching—surgically replacing the hidden state at specific positions and measuring the effect on the model’s output.

We focus on level NT5 as the target because it occupies a structural sweet spot in the hierarchy. An NT5 constituent spans several terminals (roughly 4–9): deep enough to be a coherent sub-structure, yet high enough that corrupting it breaks the *global* parse structure (CYK validity) while leaving the *local* surface statistics (n-gram texture) largely intact. This structure-vs-surface dissociation is exactly what separates “the information is used” from “we perturbed the model and it got worse.”

Both higher and lower levels lose this dissociation. A higher level such as NT2 dominates almost the entire string, so corrupting it collapses the output wholesale, leaving little local structure to preserve. NT6, one level above the terminals, is the opposite failure: an NT6 constituent expands to only 2–3 terminals, so NT6 essentially *is* the local bigram structure. Its boundaries are also dense ($\approx 40\%$ of positions), so the intervention—which we confine to a fixed-length prefix (Section 5.3)—would overwrite many positions for NT6 but only a handful for NT5. Patching NT6 is therefore a blunt, high-volume overwrite that moves the surface metric together with the structural one, so the two can no longer be dissociated. NT5 is the level at which a coherent structural unit can be swapped without dragging the local surface along with it.

5.2 Finding the Encoding Layer

Before patching, we need to identify the earliest layer at which NT5 information is fully encoded. We train a diagonal linear probe per layer (3,000 iterations each, evaluated on 500 fresh strings) and record boundary-position accuracy at each layer. Results are shown in Table 6.

NT5 information first reaches high accuracy at layer 5 (98.6%), with all subsequent layers above 98%. We therefore select **layer 5** as the intervention point: it is the earliest layer where the NT5 identity is reliably

Table 6: Diagonal probe accuracy at NT5 boundary positions, per layer.

Layer	NT5 accuracy
0	38.1%
1	26.3%
2	51.1%
3	85.2%
4	75.0%
5	98.6%
6	98.3%
7	99.7%
8	99.9%
9	99.8%
10	99.5%
11	99.7%

encoded, meaning any downstream layers that use this information must read it from here.

5.3 Activation Patching Experiment

Each trial starts from a *clean* CFG string x . We form the prefix $[\text{BOS}] + x_{1:50}$ (the first 50 terminals) and let the model generate a continuation autoregressively. The intervention is confined to this 50-token prefix: on every forward pass a hook at layer 5 overwrites the residual stream at the prefix’s NT5 boundary positions before the remaining blocks run. The generated continuation tokens are never patched directly—the corruption can reach them only by attending back onto the patched prefix positions—and we then score the completion with CYK. We compare three sources for the overwriting state against the unpatched clean baseline:

Donor. Hidden states copied from a different valid string (the *donor*) at the same prefix positions, chosen so the swap replaces the NT5 identity (criteria below). This tests whether swapping the NT5 label at the boundary causally alters the output structure.

Noise (boundary). Gaussian noise (zero-mean, variance matched to the residual stream) injected at the NT5 boundary positions only.

Noise (non-boundary). Gaussian noise injected at the same *number* of non-boundary positions, chosen at random. This is the control: if structure is stored at boundaries, corrupting non-boundaries should have little effect.

A donor string is accepted only if it (i) has a level-5 boundary at the *same absolute prefix index* as the clean string for at least three positions, and (ii) carries a *different* valid NT5 identity at each of those

shared positions; we patch exactly those shared positions. The same-index requirement matters because the model uses RoPE: a hidden state encodes relative position, so injecting a donor’s state from a different index would add a positional confound (it also forces the donor to be at least as long as the patched index). The different-identity requirement ensures the swap actually changes the parse—copying the same NT5 would be a near no-op. Donors are otherwise drawn from the natural CFG length distribution rather than length-matched, so that the higher-level information that unavoidably rides along (below) varies randomly across trials instead of systematically.

Patching an NT5 boundary does not swap the NT5 identity in isolation: there is one hidden vector per position, with no separate slot per level, so the donor’s vector also carries its higher-level ancestors (NT2–NT4) and its NT6 child. The two contaminants behave differently. Higher-level information varies randomly across the 500 randomly drawn donors, so it averages out and the systematic NT5 swap dominates. The NT6 identity, however, cannot be averaged away: a level-5 boundary is necessarily a level-6 boundary as well, so an NT6 child is swapped on *every* trial. We accept this as a limitation—cleanly isolating NT5 from NT6 would require overwriting only the probe’s NT5 subspace direction rather than the full vector, which we leave to future work.

We measure CYK validity of the generated output and n -gram KL drift relative to the clean baseline across 500 trials. Results are shown in Table 7.

Table 7: Activation patching results at NT5, layer 5 ($N=500$ trials). KL columns are the n -gram KL of the generated terminals against the grammar’s true n -gram distribution; lower is closer to grammatical surface texture.

Condition	CYK		n -gram KL		
	valid	drop	2	3	4
Clean	98.8%	—	0.0000	0.0000	0.0001
Donor	41.2%	57.6%	0.0001	0.0001	0.0001
Noise (boundary)	69.8%	29.0%	0.0000	0.0000	0.0001
Noise (non-boundary)	94.0%	4.8%	0.0000	0.0013	0.0019

5.4 Discussion

The donor condition produces a 57.6% drop in CYK validity, confirming that the NT5 identity stored at boundary positions is causally *used* during generation: replacing it with a structurally incompatible—but individually valid—NT5 label drives the model to complete the string ungrammatically. The noise-

at-boundary condition causes a smaller but still substantial drop (29.0%), confirming that these positions carry load-bearing information; the larger donor effect indicates the collapse is driven by the *identity* swap, not merely by perturbing those positions. Because the donor supplies a coherent, valid alternative parse rather than random corruption, the model reads a plausible-looking signal and follows it confidently—it is, in effect, “tricked” into committing to a competing structure that conflicts with the unpatched context—which is precisely why the donor breaks CYK more sharply than noise does.

Noise injected at *non-boundary* positions causes almost no structural disruption (4.8% drop, against a 98.8% clean baseline). This asymmetry—a 29.0% drop when the same noise hits boundaries versus 4.8% when it hits non-boundaries—is strong causal evidence for the original paper’s hypothesis: structural information is not spread uniformly across positions but concentrated at constituent boundaries, from which downstream layers read it to guide generation.

The n -gram KL columns test whether these interventions break *structure* specifically or simply degrade the output. Under the donor patch, the model’s bi-gram texture stays at the clean baseline (0.0001 vs. 0.0000) even as CYK validity collapses: the model keeps emitting locally well-formed token transitions while failing to close the global parse. This structure-vs-surface dissociation is what makes the result a *use* test rather than “we perturbed the model and it got worse.” We caution, however, that the surface axis is weakly resolved here: all n -gram KL values lie within 0.002 of zero, so there is little dynamic range on these short continuations. We had expected the donor’s KL to climb with n -gram order (local texture intact, longer-range structure breaking); instead it stays flat across orders, and the only appreciable movement is in the non-boundary noise condition at orders 3–4 (0.0013 and 0.0019). We do not over-interpret this small, prediction-contrary drift; cleanly resolving the surface axis would require longer continuations and a larger sample.

6 Discussion

Our results confirm the central qualitative findings of Allen-Zhu and Li [1]: after training on next-token prediction over CFG-generated strings, a GPT-2 model does develop internal representations that are sensitive to parse-tree structure. The attention analysis (R6–R9) is particularly striking—the DP-like attention pattern at constituent boundaries emerges clearly even at 6,500 training steps.

The high multinomial completion accuracy (98.00%) is consistent with the relatively small KL divergence (0.000350 nats/token): the model has already learned to place most of its probability mass over valid continuations. The remaining KL gap relative to the paper’s target (≈ 0.00009) reflects that the model has not yet fully converged to the exact grammar distribution, but the bulk of the structural learning has already occurred.

7 Future Work

We note first that further training is not a priority for this project. Our goal is to *reproduce* the qualitative findings of Allen-Zhu and Li [1] rather than to ship a converged model, and the structural phenomena we set out to verify—the DP-like boundary attention, the decodability of parse-tree symbols, and the causal use of NT5 boundaries—are already clearly present at 6,500 steps. Closing the residual KL gap to the paper’s target would mostly confirm convergence we already expect; it would not change any of the conclusions above. We therefore see the more interesting directions in the extension itself, where two issues are worth pursuing.

Cleaner isolation of NT5.. Our patch overwrites the full residual vector at a boundary, which unavoidably swaps the NT5 child’s NT6 constituent on every trial (Section 5.3). Averaging over random donors removes the higher-level contamination but not this nested NT6 swap, so the donor effect mixes “NT5 was changed” with “its NT6 child was changed.” A cleaner intervention would overwrite only the probe’s NT5 subspace direction—projecting the donor’s contribution onto the NT5 readout and leaving the orthogonal complement (including the NT6 signal) untouched—so the swap targets NT5 alone.

Better-resolved local-structure measurement..

The n -gram KL columns were meant to separate a broken *global* parse from intact *local* surface texture, but the surface axis is weakly resolved on our short continuations: all values sit within 0.002 of zero and the donor’s KL does not grow with n -gram order as predicted (Table 7). A sharper version of this experiment would use longer continuations and a larger trial count to give the surface metric real dynamic range, and could compare against the grammar’s true n -gram distribution at several orders to trace exactly where, in n , the corruption transitions from local to structural.

8 Conclusion

We have reproduced the experimental setup of Allen-Zhu and Li [1] on the `cfg3b` grammar using a GPT-2 model trained for 6,500 steps. Our results confirm the paper’s qualitative findings—particularly the DP-like attention pattern at constituent boundaries and the sensitivity of linear probes to parse-tree structure—while a remaining gap in KL divergence reflects incomplete convergence at our reduced training budget. The codebase is available at https://github.com/annapatata/llm_physics.

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References

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